

Gaussian Naive Bayes — Notes

1. Why Gaussian Naive Bayes?

Normal Naive Bayes works well with **categorical data**.

Example:

Weather = Sunny

Color = Red

But what if the feature is numerical?

Height = 175 cm

Weight = 70 kg

Age = 22

We cannot simply count how many times 175 appears.

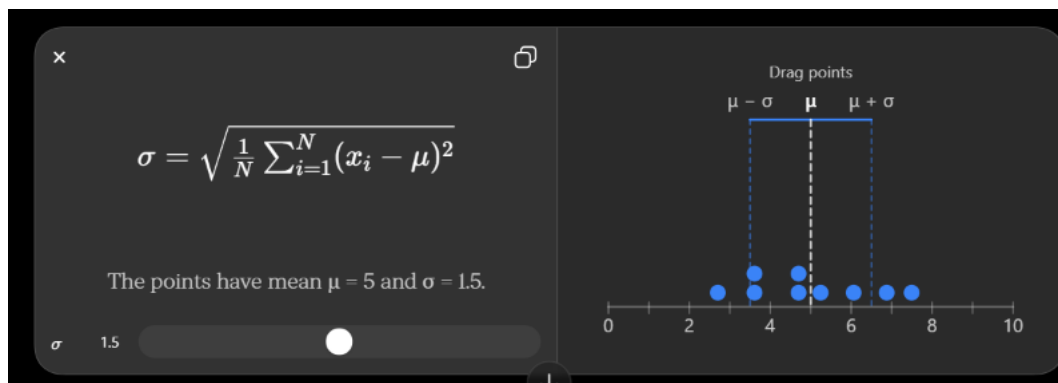
So we use **Gaussian Naive Bayes**.

2. Main Idea

Gaussian Naive Bayes assumes that the numerical feature values **follow a normal (Gaussian) distribution within each class**.

For every feature and every class, we calculate:

- **Mean (μ)**
- **Standard deviation (σ)**



3. Example

Suppose we want to classify a person as:

Class 0 → Not selected

Class 1 → Selected

Feature:

CGPA

Training data gives:

Class 0:

Mean = 6.5

Std = 0.5

Class 1:

Mean = 8.0

Std = 0.4

New student:

CGPA = 7.8

The model calculates:

How likely is CGPA = 7.8 under Class 0?

and

How likely is CGPA = 7.8 under Class 1?

It uses the **Gaussian Probability Density Function (PDF)**.

4. Gaussian PDF

$$P(x | C) = \frac{1}{\sqrt{2\pi\sigma^2}} e^{-\frac{(x-\mu)^2}{2\sigma^2}}$$

where:

- x = new value
 - μ = mean
 - σ = standard deviation
 - C = class
-

5. Final Prediction

The model combines:

Feature probabilities × Class prior probability

For example:

Class 0 → 0.10

Class 1 → 0.75

Since Class 1 has the higher probability:

Prediction → Class 1

 **Remember**

Numerical data



Calculate Mean + Standard Deviation



Gaussian PDF



Calculate probability for each class



Choose highest probability

Gaussian Naive Bayes = Naive Bayes for continuous numerical data using a Gaussian distribution.